

Competitive Exam Practice 6 — Answer Key

Hardwired Control Unit Design

PCC CS-401: Computer Organization and Architecture

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- Q1 (C).** Every hardwired control signal is a Boolean function of an opcode line D_i and a step line T_j together — neither alone identifies both “which instruction” and “which step.”
- Q2 5.** $\lceil \log_2 25 \rceil = 5$ (4 bits address only 16 codes, insufficient for 25; 5 bits address 32).
- Q3 (C).** SC counts through control-step numbers and the step decoder expands the count into one-hot T_i lines, exactly as a modulo- n counter plus decoder would.
- Q4 64.** $2^6 = 64$.
- Q5 (C).** Hardwired control is *faster* per step than microprogrammed control (pure combinational logic, no memory read) — this is an *advantage* of hardwired control, not a disadvantage, making it the correct choice for “NOT a disadvantage.”
- Q6 (B).** The OR-of-products pattern exists precisely because a control signal can be needed by more than one instruction at the same step number; each instruction that needs it contributes its own $(D_i \cdot T_j)$ term to the same shared equation.